

Measuring Light- Radiometry & Photometry

Radiometry & Photometry

- **Radiometry** is the science of **detecting and measuring light** in any **portion** of the **electromagnetic spectrum**.
- In practice, the term is usually limited to the measurement of **infrared, visible, and ultraviolet light using optical instruments**.
- **Photometry**: The branch of optics which deals with the **measurement of visible light** is known as ‘Photometry’. (‘Photo’ means light and ‘metry’ means measurement).

Photometric versus radiometric quantities

- There are two parallel systems of quantities known as photometric and radiometric quantities.
- Every quantity in one system has an analogous quantity in the other system.
- Some examples of parallel quantities include:
- Luminous flux (photometric) and radiant flux (radiometric)
- Luminance (photometric) and radiance (radiometric)
- Luminous intensity (photometric) and radiant intensity (radiometric)

Radiometric			Photometric		
Quantity	Symbol	Units	Quantity	Symbol	Units
Radiant Power	Φ_e	W	Luminous Flux	Φ_v	lumens (lm)
Radiant Intensity	I_e	W/sr	Luminous Intensity	I_v	lm/sr
Irradiance	E_e	W/m ²	Illuminance	E_v	lm/m ²
Radiance	L_e	W/m ² -sr	Luminance	L_v	lm/m ² -sr

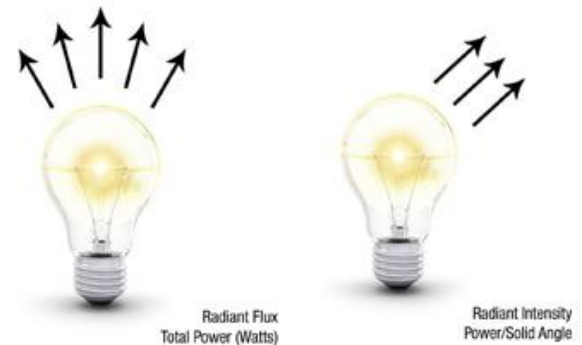
“e” = “energetic”

“v” = “visual”

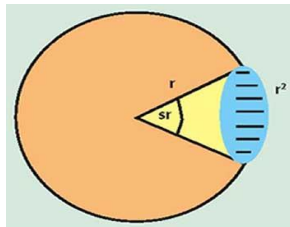
The table summarizes the most common radiometric and photometric quantities.

Radiometric quantities- definition and units

- **Radiant power** or **radiant flux** is the total **radiant energy** emitted, reflected, transmitted or received, per unit time
- It can also be defined as the rate of flow of radiant energy through a certain area or out of a certain solid angle. The SI unit of radiant flux is the Watt.
- The SI unit of radiant flux is the **watt** (W), that is joule_ per second (J/s) in SI base units



- **Radiant intensity** is the **radiant flux** emitted, reflected, transmitted or received, per unit solid angle
- The standard unit of radiant intensity is the **watt per steradian** (W/sr)
 - **solid angle** (symbol: Ω) is a measure of the amount of the field of view from some particular point that a given object covers
 - The steradian is the solid angle at the center of a sphere that subtends a surface area of r^2 .

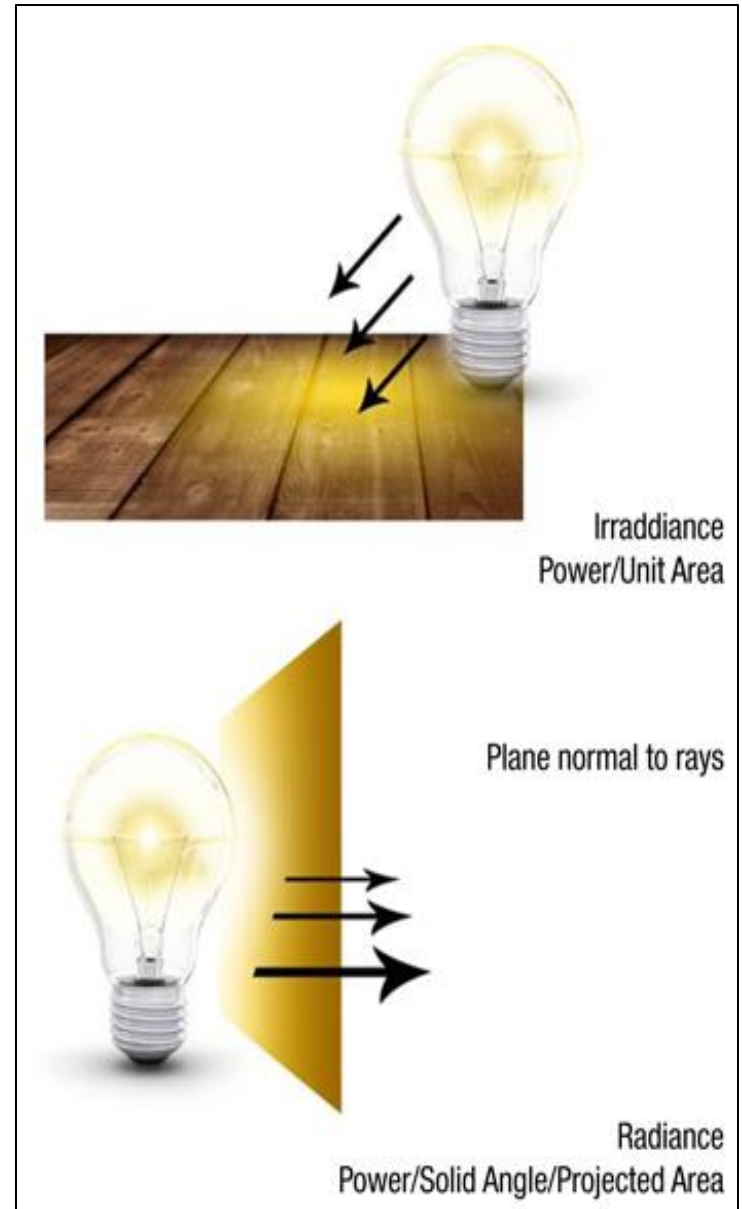


Radiant Flux
Total Power (Watts)



Radiant Intensity
Power/Solid Angle

- **Irradiance** is the **intensity of light** and is measured in **watts per square meter**.(W/m^2)
- **Radiance** : It is the Irradiance per unit solid angle
- It is often casually called “**brightness**”, a term also used in photometry to describe the perception of human eyes looking at a light source.
- The SI unit of radiance is **watts per square meter per steradian** [$\text{W}/\text{m}^2\text{-sr}$].



Photometric Quantities

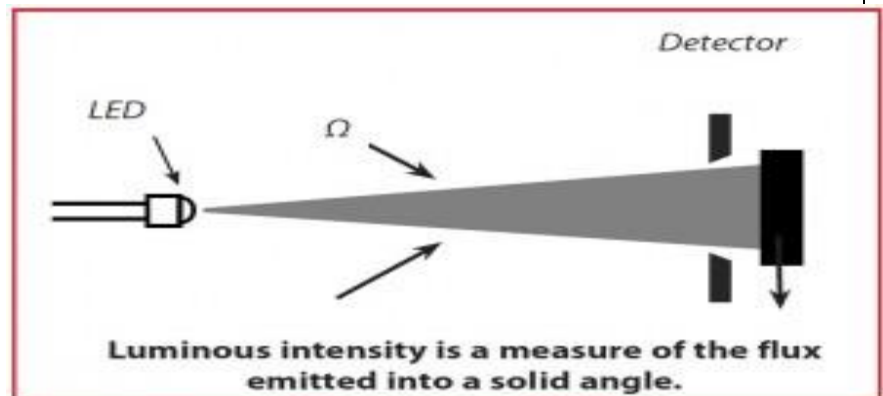
- **Luminous Flux** (Φ_v)
- A source of light radiates energy in the form of electromagnetic waves.
- If Light energy is considered as 'flux' and luminous flux is a measure of the flow of light energy emitted by a source, or received by a surface.
- The quantity is derived from the radiant flux, W (in Watts), by evaluating the radiation in accordance with the relative luminous efficiency of the 'standard eye' (CIE Standard Luminosity Function, V^*).
- The unit of luminous flux is **lumen** (lm)
- $\text{lm} = 683 \times W (\text{Watt}) \times V^*$



Luminous Flux
Total Power (lumen)
"Light Power"

- **Luminous intensity** : This expresses the power of a light source. It is defined as the **quantity of luminous flux emitted in a given direction per solid angle (in steradian)**.
- The unit is candela (cd). 1 cd = 1 lumen per steradian. (For practical purposes, one candela power.)
- the basic unit of luminous intensity is the **candela, equal to one lumen per steradian**.

(Candela is defined as the luminous intensity, in a given direction, of a source that emits monochromatic radiation of frequency 540×10^{12} Hz and that has a radiant intensity in that direction of $1/683$ watt per steradian)



- **ILLUMINANCE** :In photometry, **illuminance** is the **total luminous flux** incident on a surface, per unit area
- It is a measure of how much the incident light illuminates the surface.
- In SI derived units these are measured in **lux (lx)** or **lumens per square metre**
- The unit is lux (lx).
1 lx = 1 lumen per square meter (lm/m²)



Illuminance
Total Power/Unit Area
"Illumination"

- **LUMINANCE : Luminance** is apparent brightness, how bright an object appears to the human eye. Luminance may be thought of as luminous intensity per unit area.
- Also known as photometric brightness,
- Luminance is measured in **candelas per square meter** (cd/m^2)



- Eg-A light source illuminates a piece of paper, but your eye perceives the luminance.

Photometry and the eye

- All light measurement is considered **radiometry** with **photometry** being a special subset of radiometry weighted for a typical **human eye response**
- The human eye is **not equally sensitive to all wavelengths of visible light**.
- Photometry attempts to account for this by **weighing the measured power at each wavelength with a factor** that represents how sensitive the eye is at that wavelength.
- The standardized model of the eye's response to light as a function of wavelength is given by **the luminosity function**.
- The eye has different responses as a function of **wavelength** when it is adapted to light conditions (photopic vision) and dark conditions (scotopic vision).

- Photometry is typically based on the **eye's photopic response**, and
- so photometric measurements may not accurately indicate the perceived brightness of sources in dim lighting conditions where colors are not visible, such as under just moonlight or starlight.
- Photopic vision is characteristic of the eye's response at luminance levels over 3 candela per square metre.
- Scotopic vision occurs below $2 \times 10^{-5} \text{ cd/m}^2$.

- Photometry becomes a modern science, when Commission Internationale de l'Eclairage (CIE) met to define the response of the average human eye.
- CIE measured the light-adapted eyes of a sizeable sample group, and compile the data into the CIE Standard Luminosity Function (widely known as **photopic curve** – **chromatic perception at normal state**, and **scotopic curve** – **achromatic perception at low level of illuminance**).

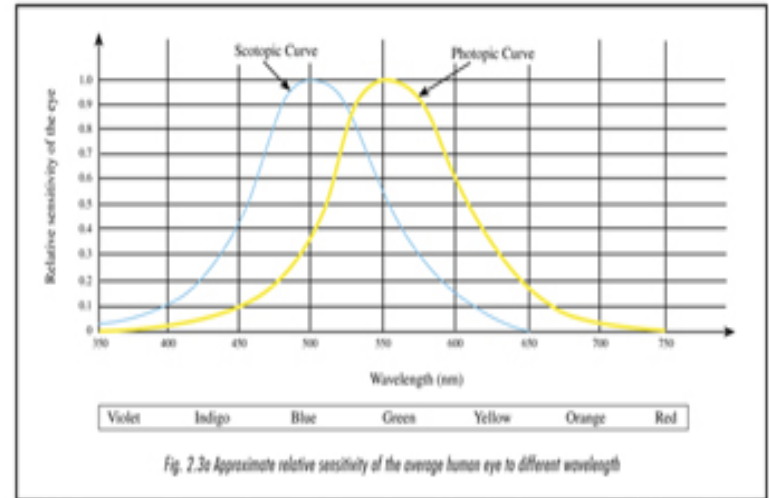


Fig. 2.3a Approximate relative sensitivity of the average human eye to different wavelength

The original curve, was adopted by the Commission on Illumination (CIE) as the standard in 1924 and is still used today even though modifications have been suggested.

Conversion - Eye sensitivity function

- The CIE 1978 $V(\lambda)$ **luminous efficiency function or eye sensitivity function**, $V(\lambda)$, which can be considered the most accurate description of the eye sensitivity in the photopic vision regime
- To convert between radiometric and photometric units, one needs to know the photopic spectral luminous efficiency curve $V(\lambda)$, which gives the spectral response of the human eye to various wavelengths of light.

- The conversion between photometric units which take into account human physiology and straight radiometric units is given by the following:
 - (photometric unit) = (radiometric unit) x (683) x $V(\lambda)$
- where $V(\lambda)$ is the 'Photopic Response,' shown earlier and basically tells us how efficiently the eye picks up certain wavelengths of light.
- The Photopic response is a function of the wavelength of light and so to convert from radiometric units to photometric units first requires knowledge of the light source.

- The photometric quantities are related to the corresponding radiometric quantities by the CIE Standard Luminosity Function.
- We can think of the luminosity function as the transfer function of a filter which approximates the behaviors of the average human eye (Fig.).

Fig. 2.3b - Relationship between radiometric units and photometric units



PHOTOMETER



- Photometric measurements are made with instruments called photometers
- It is an instrument which is used to **compare the luminous intensities of two sources.**
- A photometer is a tool used to **measure light absorption, illumination and light intensity, and there are many different types.**
- If the luminous intensity of one of the source is known, the luminous intensity of the other one can be measured

Comparison of photometry and radiometry

Photometry

- Photometry is the measurement of light, which is defined as electromagnetic radiation that is detectable by the human eye
- Is limited to the visible spectrum as defined by the response of the eye.

Radiometry

- The measurement of optical radiation, which is electromagnetic radiation within the frequency range between 3×10^{11} and 3×10^{16} Hz.
- Includes the entire optical radiation spectrum

Comparison of photometry and spectroscopy

Photometry

- Photometry measures the energy from a source using a narrow range of wavelengths.
 - Visual wavelengths from 400-700 nm
 - Narrower slice of wavelengths
- Photometry uses filters to select wavelengths.

Spectroscopy

- Spectroscopy measures energy over a wide range of wavelengths.
 - Visual spectrum
 - UV, IR spectra
 - Full EM spectra
- Spectroscopy requires instruments to get at each wavelength separately.
 - Interferometer

Applications

- **Radiometry** is a set of techniques for measuring electromagnetic radiation, including visible light.
- Radiometric techniques in optics characterize the distribution of the radiation's power in space, as opposed to photometric techniques, which characterize the light's interaction with the human eye. Radiometry is distinct from quantum techniques such as photon counting.
- The use of radiometers to determine the temperature of objects and gasses by measuring radiation flux is called pyrometry.
- Handheld pyrometer devices are often marketed as infrared thermometers.

Applications of photometry

The applications of photometry are

- Luminance meters are used in several industries to test the
 - brightness of displays
 - instrument panels,
 - night-vision devices,
 - lamp sources, etc

Weber–Fechner law

- The **Weber–Fechner law** refers to two related laws in the field of psychophysics, known as **Weber's law** and **Fechner's law**.
- Both laws relate to **human perception**, more specifically the relation between the ***actual change in a physical stimulus*** and the ***perceived change***.
- This includes stimuli to all senses: vision, hearing, taste, touch, and smell.

Fechner's law

- **Fechner** was a student of Weber and named his first law in **honor of his mentor**, since it was **Weber** who had conducted the experiments needed to formulate the law
- Fechner noticed in his own studies that different individuals have different sensitivity to certain stimuli.
- For example, the **ability to perceive differences in light intensity could be related to how good that individual's vision is.**
- He also noted that the human sensitivity to stimuli changes depends on which sense is affected.
- **Fechner's law states that the subjective sensation is proportional to the logarithm of the stimulus intensity.**

Weber's law and Fechner's law

- **Weber states that, "the minimum increase of stimulus which will produce a perceptible increase of sensation is proportional to the pre-existent stimulus"**
- **while Fechner's law is an inference from Weber's law which states that the intensity of our sensation increases only as the logarithm of an increase in energy**

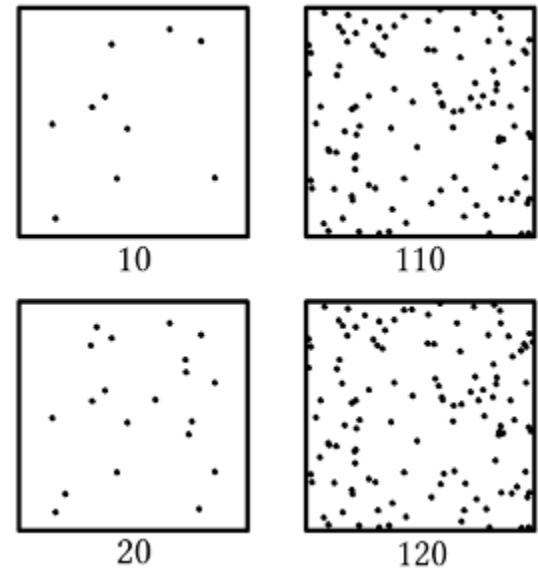
An illustration of the Weber–Fechner law.

On each side, the lower square contains 10 more dots than the upper one.

However the perception is different:

On the left side, the difference between upper and lower square is clearly visible.

On the right side, the two squares look almost the same.



Weber Fechner law Example

- **Examples:** If with one light bulb turned on in a room, we have a lighting sensation amount S , and we turn on a second equal light bulb, we will not have a double lighting sensation $2S$.

THE VISUAL DISCRIMINATION OF INTENSITY AND THE WEBER-FECHNER LAW - Differential Sensibility

- The power to distinguish differences in the brightness of objects is an outstanding property of the human eye, and is used constantly in judgments of distance and form.
- It is not possible, however, to estimate quantitatively the magnitudes of the corresponding differences in sensation.
- A light may be described as appearing slightly brighter or much brighter than another light, but no numerical values can be assigned to such judgments

- Direct estimation being impossible, it is customary to evaluate differential sensibility indirectly in terms of the minimum change in illumination which can be distinguished as a change in brightness.
- If I is the intensity to which the eye is adapted, and ΔI , is the increase in the intensity which is just perceptible, then the ratio $\Delta I / I$ may be considered a measure of the discriminating power of the eye

- The description of the relation between the discriminable threshold ΔI and the original intensity I has had an interesting evolution, culminating in what is currently known as the Weber-Fechner law
- According to this law $\Delta I / I$ is constant within wide limits of intensity
- Weber's Law, also sometimes known as the Weber-Fechner Law, suggests that the just noticeable difference is a constant proportion of the original stimulus.

Thank you